

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

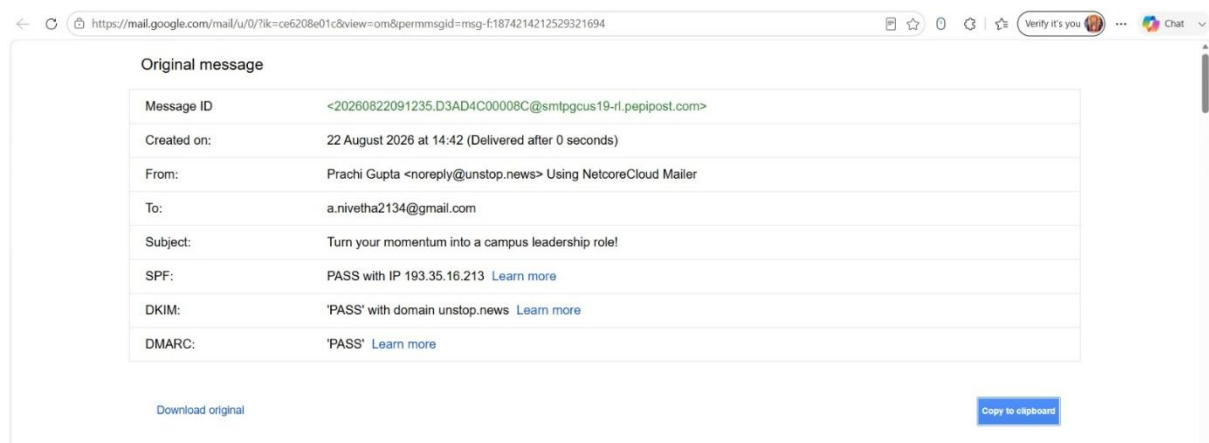
Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.



Google Admin Toolbox Messageheader

Help

MessageId

20260822091235.D3AD4C00008C@smtpgcs19-rl.pepipost.com

Created at:

8/22/2026, 2:42:37 PM GMT+5:30 (Delivered after)

From:

Prachi Gupta <noreply@unstop.news> Using NetcoreCloud Mailer

To:

a.nivetha2134@gmail.com

Subject:

Turn your momentum into a campus leadership role!

DKIM:

pass with domain unstop.news
[Learn more](#)

#	Delay	From *	To *	Protocol	Time received
0		mta-16.213.unstop.news.	→ [Google] mx.google.com	ESMTPS	8/22/2026, 2:42:37 PM GMT+5:30
1			→ [Google] 2002:a05:620a:a40a:20b0:937:2fdf:2915	SMTP	8/22/2026, 2:42:37 PM GMT+5:30
2			→ [Google] 2002:a5d:6d43:0:b0:482:bd73:c392	SMTP	8/22/2026, 2:42:37 PM GMT+5:30

Original message

Message ID

<b201fd577f8aa9486779d03ae56f789dc166d7eb-20394068-111847862@google.com>

Created on:

7 August 2026 at 01:24 (Delivered after 0 seconds)

From:

Kaggle <no-reply@kaggle.com>

To:

a.nivetha2134@gmail.com

Subject:

Competition Launch: RSNA Knee Abnormality Detection

SPF:

PASS with IP 209.85.220.69 [Learn more](#)

DKIM:

'PASS' with domain kaggle.com [Learn more](#)

DMARC:

'PASS' [Learn more](#)

Download original

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Delivered-To: a.nivetha2134@gmail.com

Received: by 2002:a05:6000:4108:b0:47:fb59:2b50 with SMTP id eh8csp667413wrb; Thu, 6 Aug 2026 12:54:18 -0700 (PDT)

X-Received: by 2002:a05:690e:12c8:b0:66a:c0fb:esb7 with SMTP id 956f58d0204a3-66ac2fae314mr2142595d50.28.1786046057980; Thu, 06 Aug 2026 12:54:17 -0700 (PDT)

ARC-Seal: i=1; a=rsa-sha256; t=1786046057; cv=none; d=google.com; s=arc-20260327; b=HfNfhjYbEjpd+Z4YCEZmK0pzyXgplbaft1b7X7jvLmfKSpH21M6G/ge7Xhnp/B VIfdovp1C7kjt3w001gG71o7pvYfac5TOFrfa0spD7QmIDr1nkdpn3HlQ1m1trUq fGA38fUdt1BxVhw351DYD5whB69jcaPrrBAVB5okAA1rshyUBOpqormszK5q6nNPQ2d9 2rAvKR88p8p8hP1zVnqFQ05EL5T/+PDawchLihpw2thzo0lQrDCGZk+c3hdIQd1p9ua JHEu2GH1ESjn1oadVd/uaM9VON08230TA0ZVU1DOFQabiJYv3en9bb1hj0Zw6lIgy Q07w==

ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=to:from:subject:message-id:mail-list-id:feedback-

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Google Admin Toolbox Messageheader

Help

MessageId

b201fd577f8aa9486779d03ae56f789dc166d7eb-20394068-111847862@google.com

Created at:

8/7/2026, 1:24:17 AM GMT+5:30 (Delivered after 1 sec)

From:

Kaggle <no-reply@kaggle.com>

To:

a.nivetha2134@gmail.com

Subject:

Competition Launch: RSNA Knee Abnormality Detection

DKIM:

pass with domain kaggle.com
[Learn more](#)

#	Delay	From *	To *	Protocol	Time received
0			→ 2002:a05:690e:4853:b0:662:c4c4:dabf	SMTP	8/7/2026, 1:24:17 AM GMT+5:30
1		mail-sor169.google.com.	→ [Google] mx.google.com		8/7/2026, 1:24:17 AM GMT+5:30
2			→ [Google] 2002:a05:690e:12c8:b0:66a:c0fb:e5b7	SMTP	8/7/2026, 1:24:17 AM GMT+5:30
3	1 sec		→ [Google] 2002:a05:6000:4108:b0:47f:fb59:2b50	SMTP	8/7/2026, 1:24:18 AM GMT+5:30

29°C Mostly cloudy

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